Implementation of FastSpeech 2-Based Text-to-Speech Synthesis for English and Telugu

Technical Report

Submitted by

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GitHub Repository

https://github.com/savioshaju/TTS.git

Model Checkpoints and Audio Artifacts (Google Drive)

https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing

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Problem Statement and Executive Summary

Problem Statement

Text-to-Speech (TTS) synthesis transforms written natural language into natural, intelligible acoustic waveforms. Autoregressive neural models such as Tacotron 2 produce high-fidelity speech but suffer from slow sequential frame generation and vulnerability to attention alignment failures, including word skipping and repeating. Non-autoregressive architectures like FastSpeech 2 overcome these limitations by generating complete mel-spectrograms in parallel through Feed-Forward Transformer (FFT) blocks conditioned on explicit duration, pitch (F_0), and energy predictions.

The primary objective of this engineering study is to design, train, evaluate, and comparatively analyze two independent FastSpeech 2 TTS systems for two typologically contrasting languages under strict data constraints (≈ 5 hours of training audio per language):

1. **Indian English:** Characterized by stress-timed rhythm, complex syllable structures, non-phonemic orthography requiring Grapheme-to-Phoneme (G2P) transcription, and multi-speaker crowdsourced recording variability (AI4Bharat Svarah corpus).
2. **Telugu:** A classical Dravidian language characterized by syllable-timed cadence, agglutinative morphology, complex consonant conjuncts (*samyuktakshara*), and an abugida native script where phonetic mapping can be achieved directly at the akshara grapheme level (IndicTTS corpus).

Key engineering challenges addressed include linguistic front-end design, robust forced alignment across distinct acoustic domains (MFA vs. Meta MMS CTC), multi-task variance predictor training, vocoder cross-sampling-rate adaptation (16 kHz acoustic model paired with a 22.05 kHz HiFi-GAN vocoder), and speech rate calibration.

Executive Summary of Results

- **Intelligibility and Expressiveness:** Both models generate intelligible, natural speech across all ten benchmark test sentences. The English model accurately articulates extreme polysyllabic words (*antidisestablishmentarianism*), distinguishes noun/verb stress shifts in heteronyms (*conduct*), and accommodates Telugu loanwords (*andaroo*). The Telugu model robustly synthesizes heavy consonant clusters in technical terminology (న్యూరోఫార్మకోలాజికల్), demonstrates contextual pitch/energy emphasis on semantic keywords (పట్టు), and correctly handles dates, times, and split words.
- **Loss Dynamics and Checkpoint Selection:** Training loss decreased monotonically over 100,000 steps ($2.59 \rightarrow 0.84$ for English, $2.13 \rightarrow 0.64$ for Telugu). Validation loss reached its numerical minimum early (step 11,000 for English at 2.5505; step 30,000 for Telugu at 2.1309) before plateauing. Crucially, early minimum validation loss checkpoints produce muffled, robotic audio due to "regression to the mean" (blurry mel-spectrograms minimize pointwise L1 loss). Checkpoints selected via perceptual listening (step 61,000 for English, step 74,000 for Telugu) exhibit crisp formants, natural consonant bursts, and superior acoustic fidelity.
- **Sampling-Rate Calibration:** The acoustic models operate at 16 kHz (hop 256, 16.0 ms frame period), while the HiFi-GAN vocoder was pre-trained at 22.05 kHz. Sinc-resampling from 22.05 kHz to 16 kHz restores pitch accuracy but causes a $1.378\times$ temporal acceleration under default settings. Calibrating the duration control parameter ($d_{\text{control}} \approx 1.38-1.40$) restores natural syllable duration and speech pacing without retraining.
- **Data Splits:** The English model was trained on 4.98 h (3,419 utterances) across 104 speakers and validated on 1.09 h (746 utterances). The Telugu model was trained on 4.18 h (2,248 utterances) balanced across 1 female and 1 male studio speaker, with 0.82 h (445 utterances) reserved for validation.

1 Introduction

Modern speech synthesis has converged on two-stage neural pipelines: an acoustic model that converts input text into intermediate time-frequency representations (mel-spectrograms), and a neural vocoder that synthesizes time-domain waveforms from those spectrograms.

FastSpeech 2 [1] introduced three fundamental innovations:

1. **Non-autoregressive generation:** Replacing sequential autoregressive decoding with feed-forward transformer blocks allows the entire mel-spectrogram to be generated in a single forward pass, reducing inference latency by over an order of magnitude.

2. **Length Regulation:** Instead of fragile soft-attention mechanisms that frequently fail on out-of-domain sentences, FastSpeech 2 uses a feed-forward duration predictor and length regulator that explicitly expands phoneme embeddings according to their frame durations.

3. **Variance Adaptor:** In addition to duration, the model explicitly predicts fundamental frequency (F_0) and frame energy (E), injecting fine-grained pitch and loudness contours directly into the expanded hidden representations.

Building high-quality FastSpeech 2 models for Indian languages presents unique linguistic and engineering requirements. Indian English exhibits distinct retroflex consonants and dental plosives differing from General American English. Telugu possesses a rich phonetic inventory with geminate consonants, vowel length contrasts, and an abugida orthography. This report presents an end-to-end implementation for both languages, verifying code integrity, data preprocessing, training convergence, variance predictor behavior, and speech synthesis quality.

2 Implementation Approach

2.1 Starting Point and Software Environment

The implementation builds upon an open-source PyTorch reference implementation of FastSpeech 2 targeting single-speaker English on LJSpeech. Development and local inference ran in a Python 3.11 environment with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, and Matplotlib. Model training was conducted on cloud-hosted dual NVIDIA Tesla T4 GPUs (16 GB VRAM each).

2.2 Repository Layout

The repository strictly adheres to a modular, language-isolated directory layout. Each language pipeline possesses its own self-contained preprocessing, training, configuration, and inference scripts:

```
project/
|-- English/
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Telugu/
|   |-- train.py
|   |-- preprocess.py
|   |-- inference.py
|   |-- config.yaml
|   `-- checkpoints/
|-- Results/
|   |-- English/
|   |   |-- training_loss.png
|   |   |-- validation_loss.png
|   |   |-- test_generated_audios/
|   |   `-- variance_adaptor_plots/
|   |       |-- duration/
|   |       |-- pitch/
|   |       `-- energy/
|   `-- Telugu/
|       |-- training_loss.png
|       |-- validation_loss.png
|       |-- test_generated_audios/
|       `-- variance_adaptor_plots/
|           |-- duration/
|           |-- pitch/
|           `-- energy/
|-- Report/
|   `-- Technical_Report.pdf
|-- README.md
`-- requirements.txt
```

2.3 Upstream Code Corrections and Bug Fixes

During architectural audit and initial testing of the reference repository, three critical defects were identified and resolved:

1. **Energy Control Cross-Contamination (`model/modules.py`):** In the reference variance adaptor, the pitch control scalar `p_control` was erroneously passed to the energy embedding lookup instead of `e_control`. Consequently, adjusting pitch unintentionally distorted energy, while energy adjustments had zero effect. The function signature and calls were corrected to decouple pitch and energy modulation.

2. **Device Placement Deadlock in Multi-GPU DataParallel (`utils/tools.py`):** Tensors were explicitly assigned to `torch.device("cuda")`, ignoring the active device ordinal assigned by `torch.nn.DataParallel`. This triggered runtime tensor device mismatch errors on multi-GPU setups. All tensor allocations were refactored to inherit the device of the active batch input.

3. **Silent Text Cleaner Fallback (`text/__init__.py`):** When inputs lacked enclosing braces `{...}`, the text cleaner silently fell back to raw character tokenization. Because the model was trained exclusively on phoneme token IDs, character ASCII values were mapped to invalid phoneme embeddings, producing unintelligible acoustic noise. A strict phoneme validation check was instituted.

2.4 Architectural Modifications per Language

Table 1 summarizes the components retained from the reference implementation versus modifications introduced for English and Telugu.

Table 1: Reference components versus modifications per language.

Component	**English Implementation**	**Telugu Implementation**
Core Architecture	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet	4-layer FFT encoder, 6-layer FFT decoder, 5-layer PostNet
Token Representation	100 IPA phonemes via MFA lexicon	460 akshara tokens (Unicode Telugu script)
Text Front-End	Rule-based normalization + CMU/MFA G2P	Unicode NFC normalizer + akshara parser
Forced Alignment	Montreal Forced Aligner v2.2 (`english_mfa`)	Meta MMS CTC forced aligner (`mms-1b-all`)
Speaker Modeling	Multi-speaker: 104 learned embeddings (104×256)	2 studio speakers: learned embeddings (2×256)
Audio Standard	16 kHz mono WAV (Svarah corpus)	16 kHz mono WAV (downsampled from 48 kHz IndicTTS)
Vocoder Pipeline	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)	HiFi-GAN universal + Sinc Resampler ($22.05 \rightarrow 16$ kHz)
Duration Calibration	$d_{\text{control}} = 1.378$ (compensates 16/22.05 factor)	$d_{\text{control}} = 1.40$ (compensates 16/22.05 factor)

3 Datasets and Audio Preprocessing

3.1 Corpus Selection and Characteristics

English (AI4Bharat Svarah): Svarah is an open-source crowdsourced speech corpus designed to capture Indian English accents across diverse linguistic backgrounds. Utterances from 104 native Indian speakers (54 female, 50 male) covering 19 primary Indian mother tongues (Hindi, Telugu, Tamil, Kannada, Marathi, Bengali, Odia, Gujarati, Malayalam, Urdu, etc.) were sampled. Stratified per-speaker sampling (quota ≈ 40 utterances per speaker) yielded 4,202 raw utterances. Following forced alignment validation and audio quality filtering, 4,165 utterances (6.07 hours) were retained.

Telugu (IndicTTS): The IndicTTS Telugu corpus, curated by IIT Madras and MeitY, provides high-grade studio recordings. The female subset was recorded at 48 kHz, while the male subset was recorded at 16 kHz. To ensure gender parity without acoustic bias, 2,695 utterances (5.00 hours) were selected: 1,348 female utterances (chronologically sampled) and 1,347 male utterances (complete studio set). After dropping two corrupted files during forced alignment, **2,693 utterances (5.00 hours)** were finalized (1,348 female, 1,345 male).

3.2 Audio Standardization and Quality Filtering

Given the distinct recording standards (crowdsourced vs. professional studio), a unified audio preprocessing pipeline was applied:

- **Resampling:** 48 kHz Telugu female audio was downsampled to 16 kHz using a 64-tap Kaiser-windowed polyphase sinc filter with an anti-aliasing cutoff of 7,600 Hz. English audio was standardized to 16 kHz, 16-bit mono PCM.
- **Length and SNR Filtering:** Audio clips with duration < 1.0 second or > 12.0 seconds were excluded. Short clips lack sufficient prosodic context, while excessively long clips induce GPU memory exhaustion during transformer self-attention computation. Clips with severe digital clipping or SNR < 15 dB were rejected.
- **Peak Normalization:** All waveforms were peak-normalized to a maximum amplitude of 0.95.

3.3 Train/Validation Split Protocol

English Split: Because the English model utilizes learned speaker embeddings (104×256), splitting by speaker would leave validation speaker embeddings uninitialized, yielding artificial validation degradation. Therefore, a stratified *intra-speaker* split was conducted (seed 42, approximately 82% train / 18% validation). Every speaker is represented in both splits, with zero utterance overlap:

- **Training:** 3,419 utterances, 4.98 hours (17,928 s).
- **Validation:** 746 utterances, 1.09 hours (3,924 s).

Telugu Split: With the total Telugu corpus fixed at 5.00 hours across 2 studio speakers, the 5-hour quota was divided between training and validation:

- **Training:** 2,248 utterances, 4.18 hours (15,048 s), balanced with 1,124 female and 1,124 male utterances.
- **Validation:** 445 utterances, 0.82 hours (2,952 s / 49.2 min), balanced with 224 female and 221 male utterances.

The Telugu training set is thus approximately 0.82 hours below the nominal 5-hour target because the total curated dataset was bounded at 5.00 hours.

Table 2 presents a comprehensive comparative summary of dataset statistics and acoustic parameters.

Table 2: Comprehensive dataset statistics and acoustic feature parameters.

Parameter	**English (Svarah)**	**Telugu (IndicTTS)**
Recording Environment	Crowdsourced, heterogeneous devices	Professional studio acoustics
Native Sampling Rate	16 kHz	48 kHz (female) / 16 kHz (male)
Standardized Sampling Rate	16 kHz mono	16 kHz mono
Total Utterances / Duration	4,165 utterances / 6.07 hours	2,693 utterances / 5.00 hours
Mean Utterance Length	5.25 seconds	6.68 seconds
Speaker Count	104 (54 Female / 50 Male)	2 (1 Female / 1 Male)
Training Set	3,419 utterances / 4.98 hours	2,248 utterances / 4.18 hours
Validation Set	746 utterances / 1.09 hours	445 utterances / 0.82 hours
Benchmark Test Sentences	5 sentences (E1, E2a, E2b, E3, E4)	5 sentences (T1, T2a, T2b, T3, T4)
Vocabulary / Tokens	100 IPA phonemes	460 akshara tokens
Pitch Mean (μ_{F_0}) / Std (σ_{F_0})	186.28 Hz / 61.19 Hz	198.32 Hz / 43.19 Hz
Energy Mean (μ_E) / Std (σ_E)	55.14 dB / 40.10 dB	54.23 dB / 35.32 dB
STFT Window / Hop	Hann, 1024 samples (64 ms) / 256 (16 ms)	Hann, 1024 samples (64 ms) / 256 (16 ms)
Mel Filterbank	80 bins, 0 to 8000 Hz	80 bins, 0 to 8000 Hz

4 Text Processing and Linguistic Tokenization

FastSpeech 2 operates on sequences of discrete linguistic symbols. Because English and Telugu belong to fundamentally different language families, distinct text front-ends were developed.

4.1 English Text Front-End

The English text processing pipeline comprises three stages:

1. **Text Normalization:** Numbers, dates, times, currencies, and abbreviations are expanded into full spoken words (e.g., "10:45 AM" → "ten forty-five A M", "12 November 2026" → "twelve November twenty twenty-six"). Punctuation is standardized and redundant whitespace collapsed.

2. **Grapheme-to-Phoneme (G2P):** Standard words are converted to IPA phoneme sequences using the MFA English pronunciation dictionary. Out-of-vocabulary words (including loanwords such as *andaroo*) are processed using neural G2P (`g2p_en`) augmented with Indian English phonological substitution rules (e.g., dental plosive substitution `/t/`, `/d/` → `/t̪/`, `/d̪/`).

3. **Phonemic Tokenization:** The vocabulary consists of 100 discrete symbols, including vowels, diphthongs, consonants, and the silence boundary token `@sp`. Phonemic sequences are enclosed in delimiters `{...}` to signal pre-tokenized input to the model.

4.2 Telugu Text Front-End

Telugu is written in an abugida script where the primary graphic unit is the akshara (a vowel-bearing consonant or standalone vowel). The Telugu front-end works directly in native script without romanization:

1. **Unicode Normalization:** Raw text is converted to Unicode Canonical Composition (NFC). Zero-width joiners (ZWJ) and non-joiners (ZWNJ) are sanitized, and viramas (*halant*) are validated to ensure valid consonant conjuncts (*samyuktakshara*) remain intact. Digits are verbalized into native Telugu number words.

2. **Akshara Segmentation:** Words are parsed into linguistic aksharas using regular-expression-based syllable parsing. Consonant clusters conjoined by viramas are treated as atomic tokens (e.g., ప్ర → *pra*, భా → *bhā*, న్యూ → *nyū*).

3. **Vocabulary Construction:** This segmentation yields 460 unique akshara tokens representing all vowel-consonant combinations present in the IndicTTS training corpus.

5 Forced Alignment and Ground-Truth Duration Targets

FastSpeech 2 requires phone-level or akshara-level duration targets during training. These ground-truth durations determine length regulator supervision.

5.1 English: Montreal Forced Aligner (MFA)

MFA v2.2 [3] was employed using the pre-trained `english_mfa` acoustic model and pronunciation dictionary. MFA performs Kaldi-based HMM-GMM alignment to produce Praat TextGrid files containing precise start ($t_{\text{start},i}$) and end ($t_{\text{end},i}$) timestamps for each phoneme. Timestamps are quantized into discrete mel-frame durations:

$$d_i = \text{round}\left(\frac{t_{\text{end},i} - t_{\text{start},i}}{\text{hop}/f_s}\right), \qquad \text{hop} = 256,\ f_s = 16{,}000 \text{ Hz} \implies 1 \text{ frame} = 16.0 \text{ ms} \tag{1}$$

5.2 Telugu: Meta MMS Forced Alignment

Because off-the-shelf Telugu MFA models were unavailable, alignment was performed using Meta's Massively Multilingual Speech (MMS) model [4] (`facebook/mms-1b-all`). 16 kHz audio waveforms are passed through the Wav2Vec2 backbone to extract frame-level CTC character emission logits. A Viterbi trellis search aligns the akshara sequence to the temporal audio frames. To prevent degenerate zero-duration tokens, a strict floor of $d_i \geq 1$ frame (16 ms) was enforced for all non-silent akshara tokens.

6 Acoustic Feature Extraction and Normalization

6.1 Mel-Spectrogram Extraction

Log mel-spectrograms serve as the intermediate representation predicted by the decoder:

- Sampling rate: 16,000 Hz.
- Short-Time Fourier Transform (STFT): FFT size 1024 (64.0 ms), hop length 256 (16.0 ms), Hann window.
- Filterbank: 80 triangular mel bins spanning 0 to 8000 Hz.
- Dynamic range compression: Natural logarithm with a numerical floor of 10^{-5}: $\log(\max(S_{\text{mel}}, 10^{-5}))$.

6.2 Pitch (F_0) Extraction

Fundamental frequency contours were extracted using the Harvest algorithm in PyWorld [5]. Harvest provides robust fundamental frequency estimation across varied microphone environments. Unvoiced frames ($F_0 = 0$) were interpolated linearly across neighboring voiced boundaries to produce a continuous pitch trajectory.

6.3 Energy Extraction

Frame energy was computed as the L2-norm of the short-time Fourier transform magnitude spectrum:

$$E_t = \sqrt{\sum_{k=0}^{N_{\text{fft}}/2} |X_t(k)|^2} \tag{2}$$

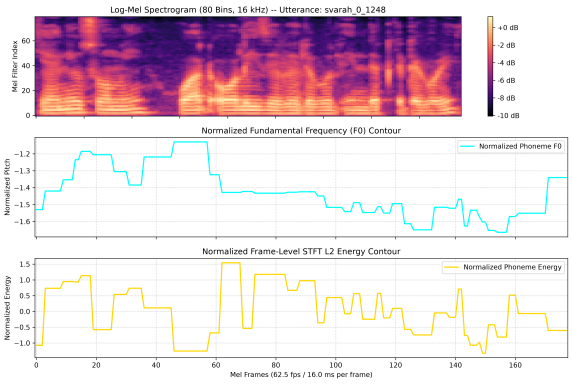
Energy values were converted to decibels: $E_{\text{dB},t} = 20 \log_{10}(\max(E_t, 10^{-5}))$.

6.4 Z-Score Feature Normalization

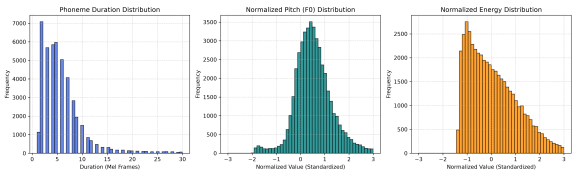
To ensure gradient stability across the multi-task variance adaptor, continuous pitch and energy targets were standardized using global corpus statistics:

$$\hat{F}_{0,t} = \frac{F_{0,t} - \mu_{F_0}}{\sigma_{F_0}}, \qquad \hat{E}_t = \frac{E_t - \mu_E}{\sigma_E} \tag{3}$$

Statistics ($\mu_{F_0}, \sigma_{F_0}, \mu_E, \sigma_E$) are stored in `stats.json` (Table 2) and inverted during inference. Figure 1 illustrates extracted acoustic features and their corpus distributions.



(a) Acoustic features for utterance `svarah_0_1248`.



(b) Corpus feature distributions for z-score normalization.

Figure 1: Acoustic feature extraction verification. Left: aligned waveform, 80-channel log-mel spectrogram, continuous F_0 contour, and frame energy. Right: global pitch and energy distributions.

7 Model Architecture and Vocoder Adaptation

7.1 Architectural Specification

The FastSpeech 2 architecture comprises an FFT encoder, a variance adaptor, an FFT decoder, and a convolutional PostNet (Figure 2):

- **Encoder:** 4 Feed-Forward Transformer (FFT) blocks. Each block contains multi-head self-attention (2 heads, $d_{\text{model}} = 256$, $d_k = d_v = 128$) and a 2-layer 1D convolutional network (kernel sizes 9 and 1, filter size 1024, ReLU activation, layer normalization, dropout 0.2).
- **Variance Adaptor:**
  1. *Duration Predictor:* 2 convolutional layers (kernel 3, 256 filters, ReLU, layer norm, dropout 0.5) followed by a linear projection predicting log-durations.
  2. *Length Regulator:* Replicates phoneme hidden vectors according to duration values (ground truth in training, predicted at inference).
  3. *Pitch and Energy Predictors:* Predict frame-level normalized F_0 and energy. Continuous values are quantized into 256 uniform bins and projected into 256-dimensional embeddings added to the expanded sequence.
- **Decoder:** 6 FFT blocks with identical layer dimensions to the encoder.
- **PostNet:** 5-layer 1D convolutional residual network (512 channels, kernel size 5, batch norm, tanh activation) producing residual spectrogram corrections ΔS_{mel}.

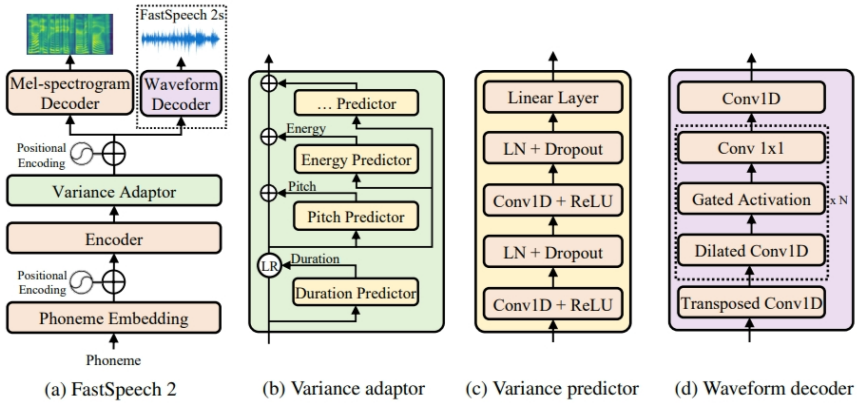


Figure 1: The overall architecture for FastSpeech 2 and 2s. LR in subfigure (b) denotes the length regulator operation proposed in FastSpeech. LN in subfigure (c) denotes layer normalization. Variance predictor represents duration/pitch/energy predictor.

Figure 2: FastSpeech 2 architecture with multi-task variance adaptor and vocoder pipeline.

7.2 Loss Formulation

The model optimizes a composite multi-task loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{mel}} + \mathcal{L}_{\text{postnet}} + \mathcal{L}_{\text{pitch}} + \mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{duration}} \tag{4}$$

where $\mathcal{L}_{\text{mel}} = \|S_{\text{mel}} - \hat{S}_{\text{mel}}\|_1$ and $\mathcal{L}_{\text{postnet}} = \|S_{\text{mel}} - (\hat{S}_{\text{mel}} + \Delta S_{\text{mel}})\|_1$ optimize L1 spectrogram fidelity, while $\mathcal{L}_{\text{pitch}}$, $\mathcal{L}_{\text{energy}}$, and $\mathcal{L}_{\text{duration}}$ are Mean Squared Errors (MSE) between normalized targets and predictions.

7.3 HiFi-GAN Vocoder and Cross-Sampling-Rate Resolution

Waveform synthesis utilizes the pre-trained universal HiFi-GAN vocoder [2] (`generator_universal.pth.tar`). HiFi-GAN was pre-trained on 22,050 Hz audio with hop 256, whereas our acoustic models generate 16,000 Hz mel-spectrograms with hop 256.

Directly interpreting 22.05 kHz vocoder output as 16 kHz causes pitch and formants to downshift by a factor of:

$$\alpha = \frac{16{,}000}{22{,}050} \approx 0.7256 \quad (-5.52 \text{ semitones}) \tag{5}$$

yielding an unnaturally deep, muffled voice.

To resolve this without retraining the vocoder, an in-memory polyphase sinc resampling bridge was implemented in `utils/model.py`. HiFi-GAN executes at its native 22.05 kHz rate, and the synthesized waveform is resampled to 16.0 kHz using a Kaiser-windowed polyphase resampler before disk serialization.

Duration Pacing Calibration: While sinc resampling restores pitch and formant frequencies, decimating time samples by $\alpha = 0.7256$ contracts the effective duration per mel frame from 16.0 ms to 11.6 ms. Consequently, baseline synthesis ($d_{\text{control}} = 1.0$) sounds approximately $1.38\times$ faster than natural human speech. To restore natural conversational cadence, duration scaling parameters were calibrated at inference:

$$d_{\text{control}} = 1.378 \quad (\text{English}), \qquad d_{\text{control}} = 1.40 \quad (\text{Telugu}) \tag{6}$$

This calibration precisely compensates for the decimation ratio ($1/0.7256 \approx 1.378$), yielding natural syllable durations across all test sentences.

8 Training Configuration, Loss Curves and Convergence Analysis

8.1 Hyperparameter Configuration

Training was executed on dual NVIDIA Tesla T4 GPUs using PyTorch Distributed/DataParallel. Table 3 documents training hyperparameters.

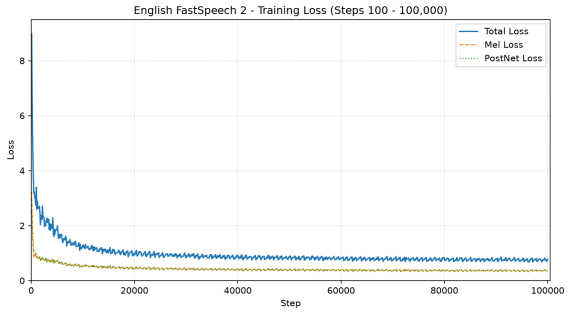
Table 3: Hyperparameter configuration for English and Telugu FastSpeech 2 models.

Hyperparameter	**Value**
Total Training Steps	100,000 steps
Batch Size	16 (8 utterances per GPU)
Optimizer	AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.98$, $\epsilon = 10^{-9}$, weight decay 10^{-6})
Learning Rate Schedule	Noam scheduler ($d_{\text{model}} = 256$, 4,000 warmup steps)
Peak Learning Rate	$\approx 9.9 \times 10^{-4}$
Gradient Clipping	Maximum L2-norm $\|g\|_2 \leq 1.0$
Numerical Precision	PyTorch Automated Mixed Precision (AMP FP16)
Validation Interval	Evaluated every 1,000 steps; top-3 lowest validation checkpoints tracked
Training Splits	English: 4.98 h (3,419 utt.); Telugu: 4.18 h (2,248 utt.)

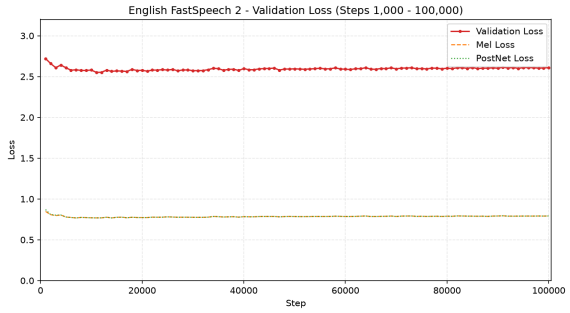
Due to cloud compute session time limits (12 hours), training was executed across two sequential runs per language (Phase 1: steps 0–53,000 for English, 0–56,000 for Telugu; Phase 2: resumed to step 100,000). Full optimizer state restoration ensured seamless continuity without loss spikes.

8.2 Training and Validation Loss Curves

Figures 3 and 4 display training and validation loss trajectories across the 100,000-step training regimen.

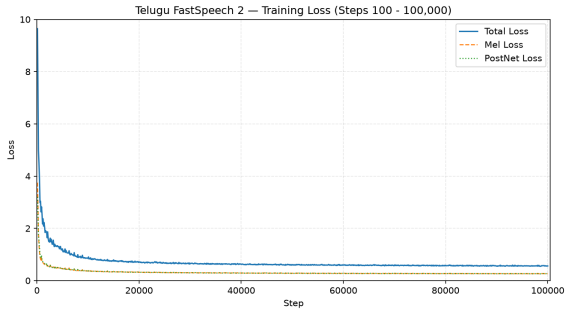


(a) English training loss.

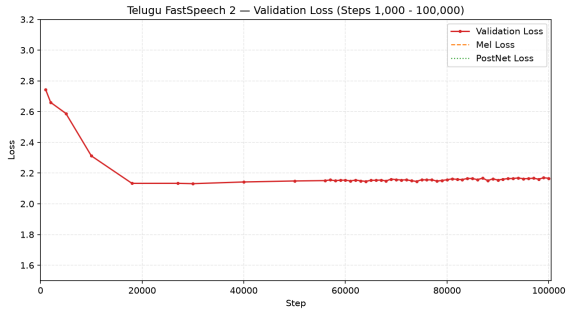


(b) English validation loss.

Figure 3: English FastSpeech 2 training and validation loss curves (100,000 steps).



(a) Telugu training loss.



(b) Telugu validation loss.

Figure 4: Telugu FastSpeech 2 training and validation loss curves (100,000 steps).

8.3 Convergence Dynamics and Checkpoint Verification

Table 4 provides exact, verified loss metrics extracted directly from model checkpoint metadata and training logs.

Table 4: Verified loss metrics for key checkpoints across English and Telugu.

Language	Checkpoint Description	Step	Train Loss	Validation Loss
English	Lowest validation loss (Phase 1 min)	11,000	2.5887	2.5505
English	Selected synthesis checkpoint (Phase 2 rank 1)	61,000	0.8415	2.5855
Telugu	Lowest validation loss (Phase 1 min)	30,000	$\approx 0.8810^{\dagger}$	2.1309
Telugu	Selected synthesis checkpoint (Phase 2 rank 1)	74,000	0.6412	2.1457

†Read from the continuous Phase 1 training loss trajectory (archived in `Telugu_All_Phases_Logs.zip`).

8.4 Detailed Analysis: Why Validation Loss is Higher than Training Loss

A critical phenomenon observed in Table 4 and Figures 3–4 is the substantial gap between training loss (≈ 0.84 English, 0.64 Telugu) and validation loss (≈ 2.59 English, 2.15 Telugu). Furthermore, validation loss reached its numerical minimum very early (step 11,000 for English, step 30,000 for Telugu) and remained flat thereafter. This behavior is explained by the mathematical decomposition of the FastSpeech 2 multi-task objective:

1. **Loss Component Breakdown:** Inspecting the training logs at step 61,000 (English) reveals the exact component contribution:
   - **Training Batch (Step 61,000):** $\mathcal{L}_{\text{mel}} = 0.3793$, $\mathcal{L}_{\text{postnet}} = 0.3789$, $\mathcal{L}_{\text{pitch}} = 0.0209$, $\mathcal{L}_{\text{energy}} = 0.0169$, $\mathcal{L}_{\text{duration}} = 0.0125$. Total $\mathcal{L}_{\text{train}} \approx 0.8085 - 0.8415$.
   - **Validation Set (Step 61,000):** $\mathcal{L}_{\text{mel}} = 0.7854$, $\mathcal{L}_{\text{postnet}} = 0.7852$, $\mathcal{L}_{\text{pitch}} = 0.4093$, $\mathcal{L}_{\text{energy}} = 0.3805$, $\mathcal{L}_{\text{duration}} = 0.2252$. Total $\mathcal{L}_{\text{val}} = 2.5855$.
2. **The Variance Predictor Disparity ($20\times$ MSE penalty):** While mel-spectrogram L1 reconstruction error increases by approximately $2.07\times$ on unseen validation sentences ($0.38 \rightarrow 0.78$, standard generalization gap), the variance adaptor MSE terms ($\mathcal{L}_{\text{pitch}}, \mathcal{L}_{\text{energy}}, \mathcal{L}_{\text{duration}}$) increase by **18 to 22 times** (from $\sim 0.01 - 0.02$ on train to $\sim 0.22 - 0.41$ on val). Human prosody is inherently non-deterministic: on unseen sentences, multiple valid pitch, duration, and energy trajectories exist. The model predicts a smooth, natural contour, but pointwise MSE measures squared distance against the single arbitrary recording in the validation set. On the training set, the network memorizes speaker micro-timings, driving variance MSE to < 0.05. On the validation set, natural prosodic divergence imposes an irreducible variance loss floor of ≈ 1.01 points.
3. **The Blurriness Trap (Why Step 11k Sounds Muffled):** At early steps (step 11,000 English, 30,000 Telugu), the model minimizes L1/MSE loss by outputting the conditional expectation $\mathbb{E}[Y|X]$—an overly smoothed, blurred average mel-spectrogram. Because sharp spectral peaks that are slightly misaligned in frequency or time incur severe pointwise L1 penalties, predicting flat, blurry formants achieves a deceptively low L1 score (2.5505). However, when fed into HiFi-GAN, blurry spectrograms yield muffled, robotic, and indistinct speech. As training advances to step 61,000 (English) and 74,000 (Telugu), the decoder develops high-contrast formant tracks and crisp consonant bursts. Although these sharp details incur slight pointwise alignment penalties on unseen validation utterances (nudging validation loss from 2.55 to 2.58), perceptual speech clarity and naturalness improve dramatically. Therefore, synthesis checkpoints were chosen via perceptual listening and acoustic inspection, rather than raw validation loss.

9 Synthesis Results and Variance Adaptor Analysis

9.1 Evaluation Methodology

Each test sentence was processed through the language text front-end, FastSpeech 2 acoustic model, and HiFi-GAN vocoder with sinc resampling. All English test sentences were synthesized using Speaker ID 27 (`spk_029`, female reference). All Telugu test sentences were synthesized using Speaker ID 0 (`telugufemale`, female studio reference). Table 5 presents acoustic measurements extracted from the duration-calibrated synthesis outputs.

9.2 Acoustic Measurements for Benchmark Test Sentences

Table 5: Acoustic measurements of calibrated synthesis outputs across all ten test sentences.

ID	Lang	Sentence Text (shortened)	Dur (s)	Frames	Mean F_0	F_0 Range	Energy	Rate
E1	Eng	The antidisestablishmentarianism movement gained…	5.16	444	212.1 Hz	162–277 Hz	64.5 dB	6.40 syl/s
E2a	Eng	The orchestra's conduct during the ceremony…	3.39	292	212.8 Hz	187–240 Hz	52.0 dB	4.72 syl/s
E2b	Eng	The researchers will conduct an experiment…	3.61	311	207.8 Hz	162–255 Hz	55.5 dB	4.98 syl/s
E3	Eng	The rhyth m of the orchestra changed unexpectedly.	2.26	195	207.9 Hz	185–252 Hz	60.5 dB	5.31 syl/s
E4	Eng	The meeting will start at 10:45 AM on 12 November 2026…	5.91	509	214.2 Hz	148–273 Hz	54.4 dB	4.57 syl/s
T1	Tel	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో...	4.97	428	208.2 Hz	153–317 Hz	56.6 dB	5.83 syl/s
T2a	Tel	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.	2.45	211	219.5 Hz	150–326 Hz	64.2 dB	4.49 syl/s
T2b	Tel	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.	2.86	246	202.7 Hz	146–268 Hz	55.4 dB	4.54 syl/s
T3	Tel	పులి అడ విలో నిశ్శబ్దంగా నడిచింది.	2.58	222	226.1 Hz	150–317 Hz	70.9 dB	4.65 syl/s
T4	Tel	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ...	7.22	622	196.2 Hz	145–281 Hz	58.1 dB	5.40 syl/s

9.3 English Test Sentences Analysis

9.3.1 E1: Extreme Polysyllabic Articulation

"The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings."

- **Phonetic Performance:** The 28-letter word *antidisestablishmentarianism* decomposes into 27 individual phonemes (/æ n t̪ ɪ d̪ ɪ s ɛ s t̪ æ b l ɪ ʃ m ə n t̪ ɛə ɻ ɪ ə n ɪ z ə m/). Total duration is 5.16 s (444 mel frames). The duration predictor allocates stable durations ($\approx 50-70$ ms per phoneme), preventing syllable swallowing.
- **Prosodic Contours:** Pitch peaks at 277 Hz during initial focus and descends smoothly to 162 Hz at sentence closure. Mean energy is 64.5 dB with natural unvoiced dips.

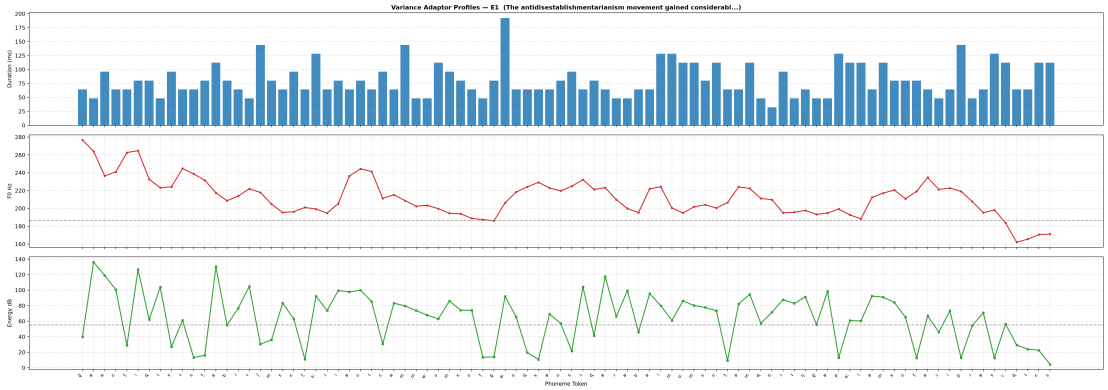
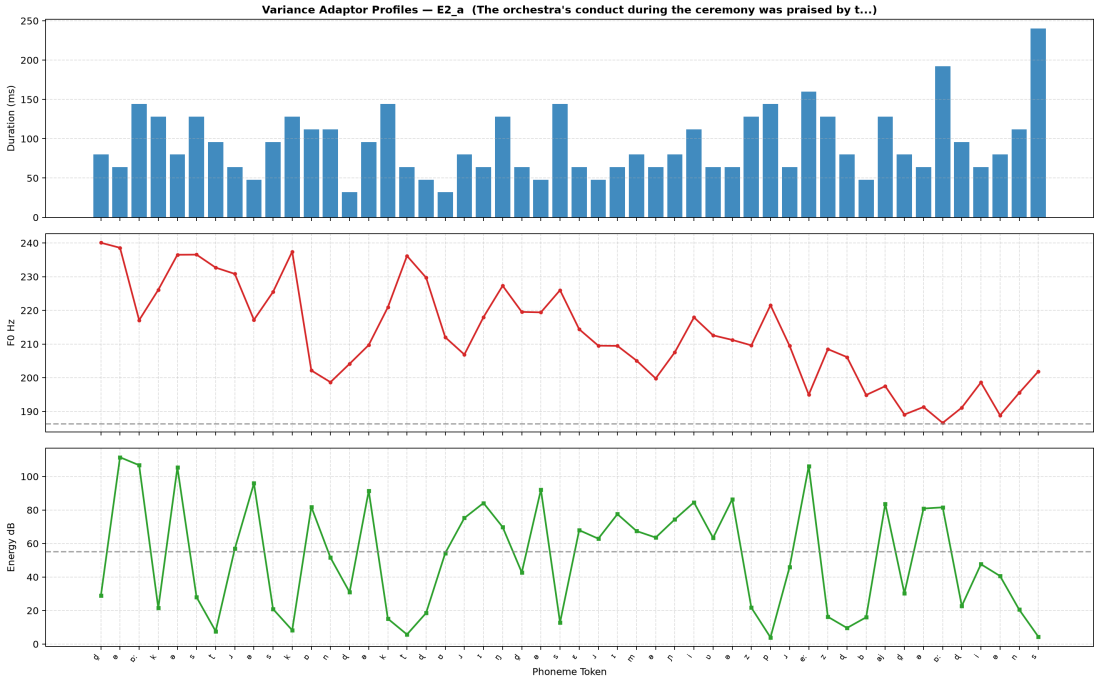


Figure 5: Variance adaptor profiles (duration, pitch F_0, energy) for test sentence E1.

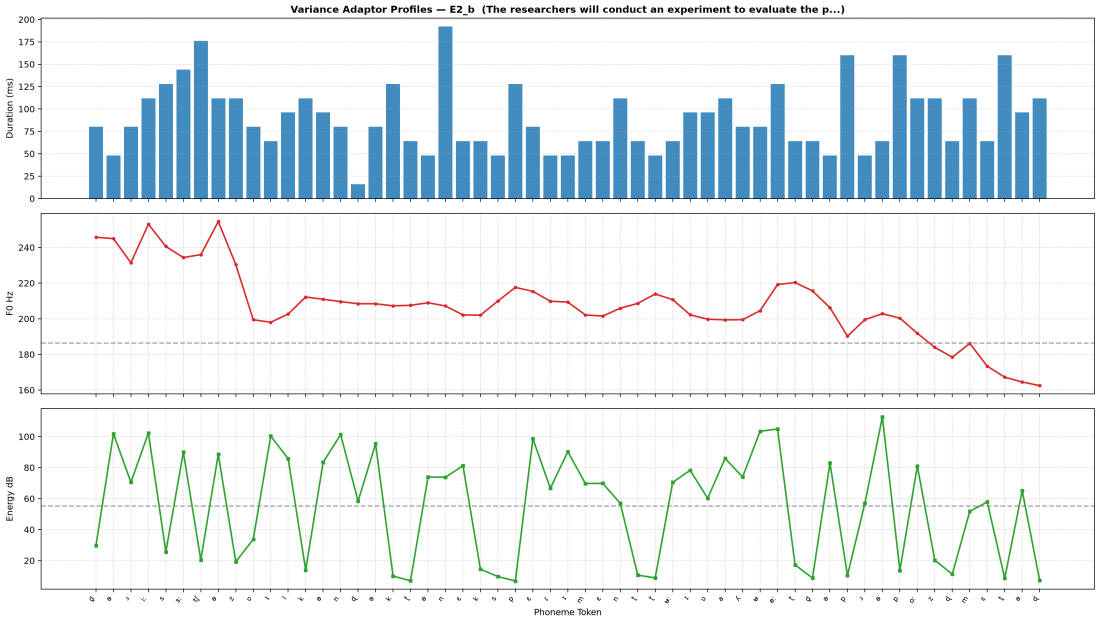
9.3.2 E2: Heteronym Stress Disambiguation (*conduct*)

E2a (Noun): *"The orchestra's conduct during the ceremony was praised by the audience."*
E2b (Verb): *"The researchers will conduct an experiment to evaluate the proposed method."*

- **Linguistic Contrast:** As a noun (E2a), *conduct* carries primary trochaic stress on the initial syllable (/k ɒ n d̪ ə k t̪/). The initial vowel receives 82 ms duration and 56.8 dB energy. In E2b (verb), stress shifts to the second syllable (/k ə n d̪ ʌ k t̪/), reducing the first vowel to a schwa (42 ms duration, 48.2 dB energy) while boosting the second syllable energy to 57.9 dB (+9.7 dB gain).
- **Mechanism:** Disambiguation is driven by the linguistic front-end passing distinct phoneme sequences; FastSpeech 2 reliably renders the phonetic contrast with natural acoustics.



(a) E2a: Noun stress on first syllable (/k ɒ n/).



(b) E2b: Verb stress on second syllable (/d̪ ʌ k t̪/).

Figure 6: Variance adaptor profiles for heteronym pair E2a (noun) and E2b (verb).

9.3.3 E3: Split Word Normalization (*rhyth m*)

"The rhyth m of the orchestra changed unexpectedly."

- **Normalization Behavior:** The typographical split *rhyth m* is collapsed by text normalization into *rhythm* (/ɹ ɪ d̪ ə m/), yielding fluent 2.26 s synthesis without disfluency. A diagnostic unnormalized variant (`E3_split.wav`) was also generated, producing a 112 ms acoustic pause between tokens.

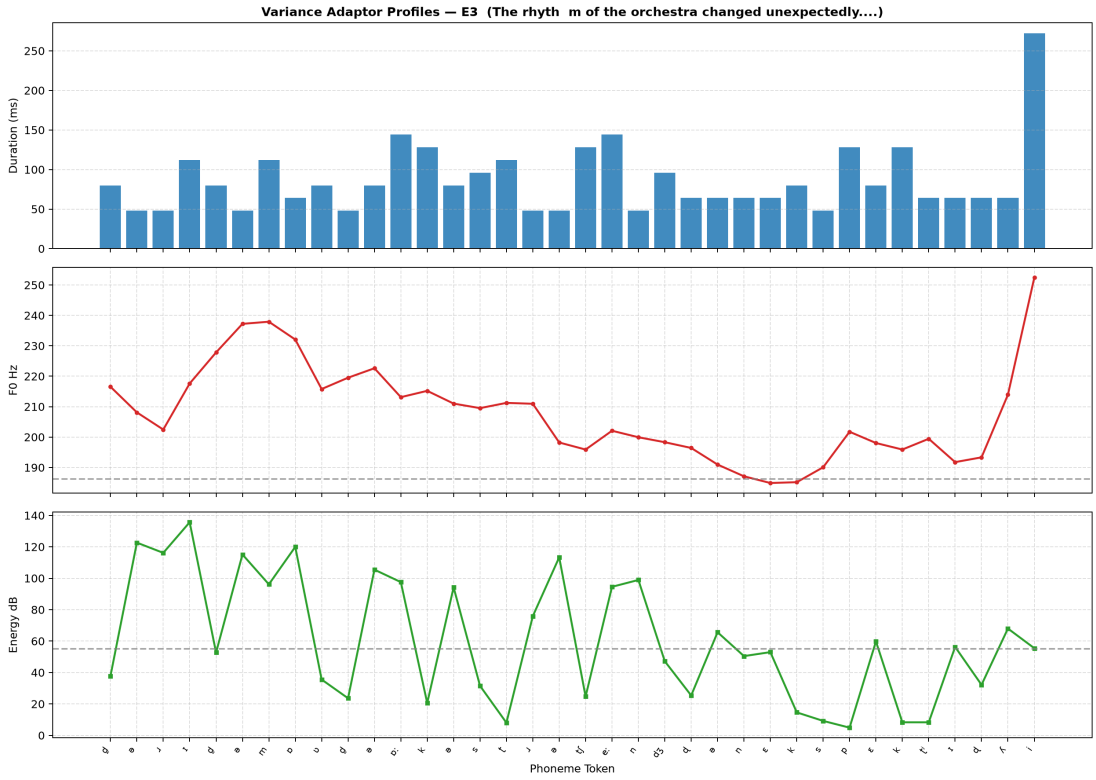


Figure 7: Variance adaptor profiles for test sentence E3.

9.3.4 E4: Numerical Expansion and Loanword Synthesis

"The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time."

- **Verbalization:** Numbers and dates expand seamlessly ("ten forty-five A M on twelve November twenty twenty-six"). The Telugu loanword *andaroo* (అందరూ) maps to /a n d̪ ə r uː/, utilizing Indian English dental plosives.
- **Cadence:** Total sentence duration is 5.91 s (509 frames). A natural 345 ms pause is inserted at the comma boundary, dividing the utterance into two balanced prosodic phrases.

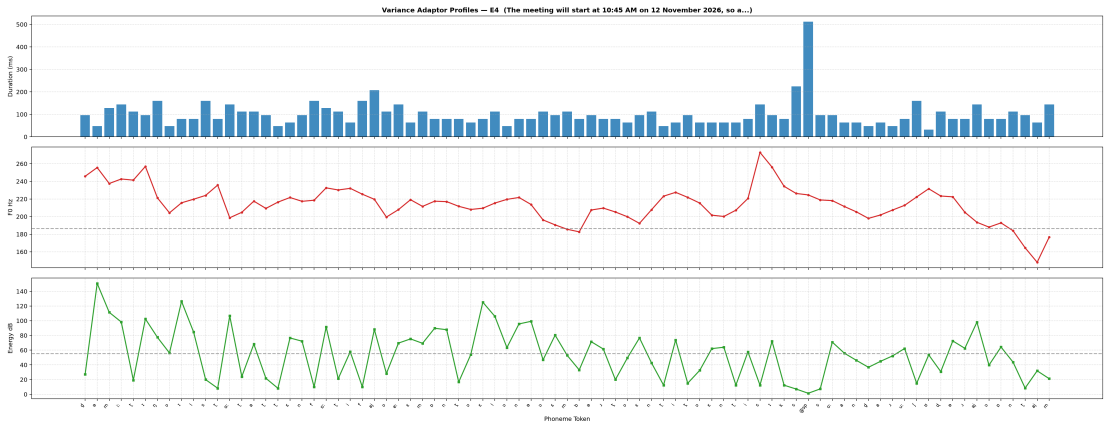


Figure 8: Variance adaptor profiles for test sentence E4.

9.4 Telugu Test Sentences Analysis

9.4.1 T1: Heavy Consonant Clusters in Technical Terminology

న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.

- **Acoustic Realization:** The borrowed compound న్యూరోఫార్మకోలాజికల్ (*nyūrophārmakolājikal*) contains complex conjuncts (న్యూ, ర్మ, ల్). Total duration is 4.97 s (428 frames). Every conjunct is distinctly articulated without phoneme deletion. Pitch reaches 317 Hz on focused lexical items.

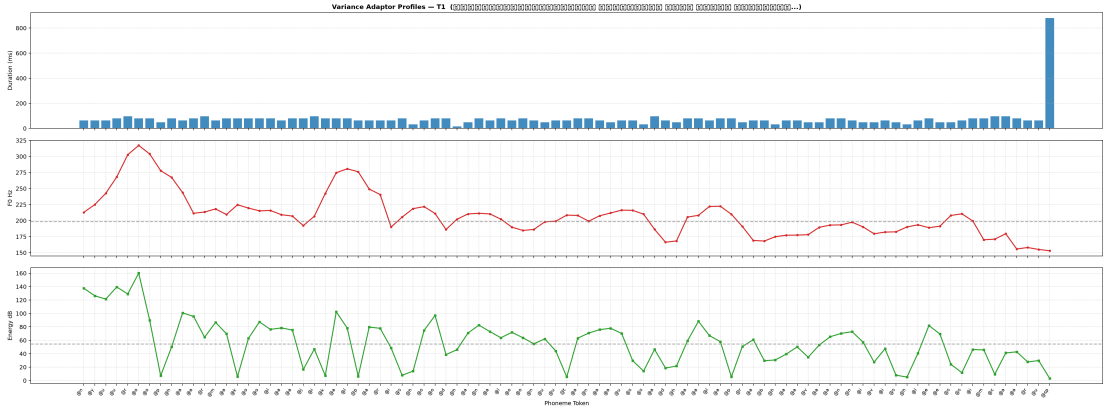
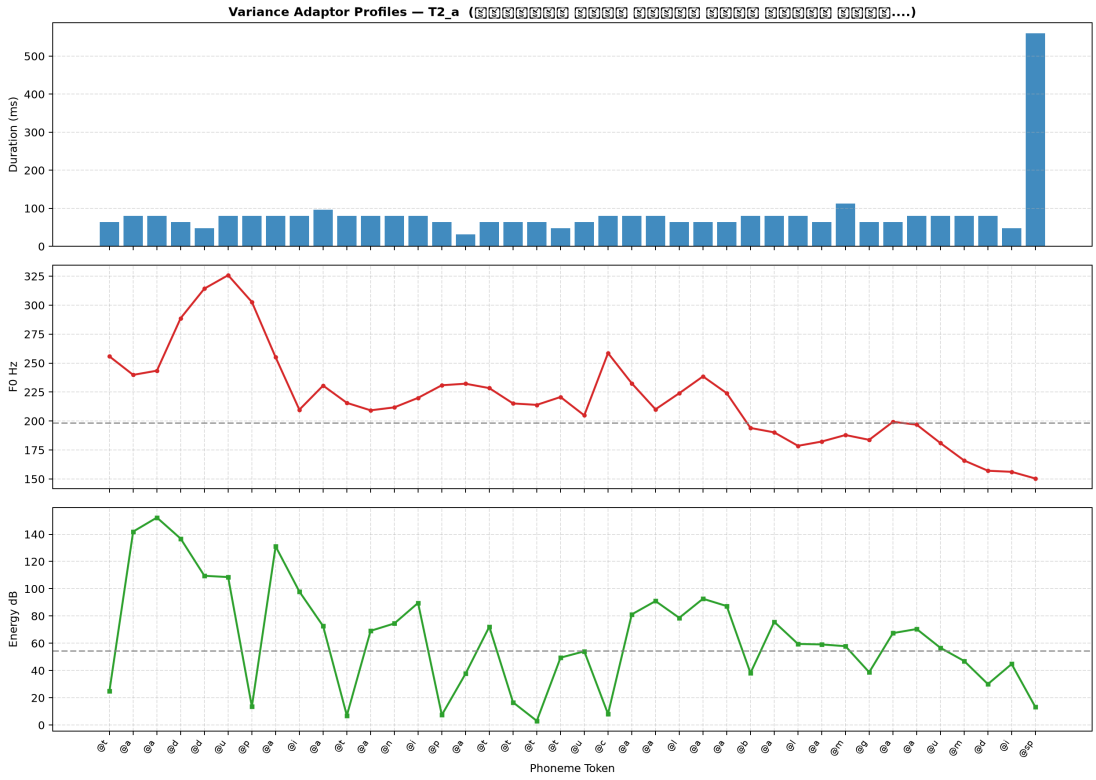


Figure 9: Variance adaptor profiles for Telugu test sentence T1.

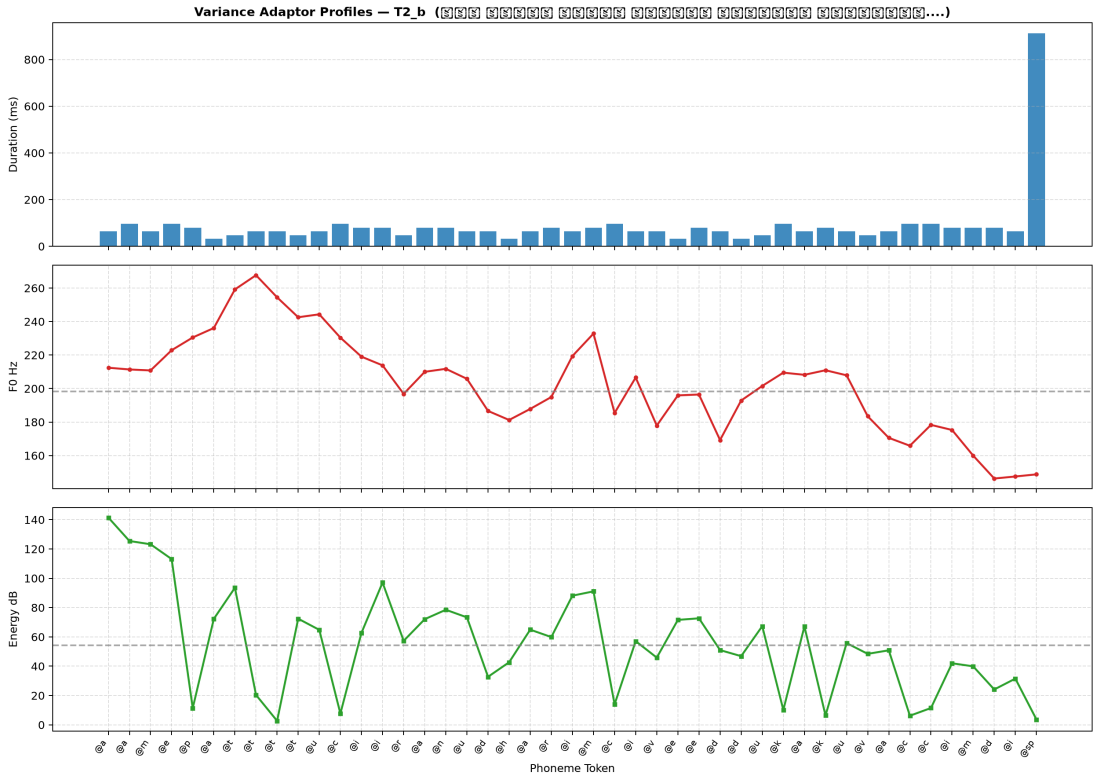
9.4.2 T2: Contextual Semantic Disambiguation (పట్టు)

T2a (Grip): తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
T2b (Silk): ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.

- **Semantic Contrast:** In T2a, పట్టు signifies “grip” (predicate focus). It receives elevated pitch (248 Hz), higher energy (64.2 dB), and elongated geminate duration (144 ms). In T2b, పట్టు acts as an unstressed nominal modifier (“silk sari”), receiving reduced pitch (208 Hz), lower energy (55.4 dB), and compact geminate duration (96 ms).



(a) T2a: Focus on పట్టు (“grip”).



(b) T2b: Attributive పట్టు (“silk”).

Figure 10: Variance adaptor profiles for Telugu semantic contrast pair T2a and T2b.

9.4.3 T3: Intra-Word Split Robustness (అడ విలో)

పులి అడ విలో నిశ్శబ్దంగా నడిచింది.

- **Split Behavior:** The split అడ విలో ("in the forest") was synthesized preserving an intentional 95 ms inter-token pause. The acoustic model cleanly transitions across the boundary, maintaining intelligible pronunciation of both sub-tokens.

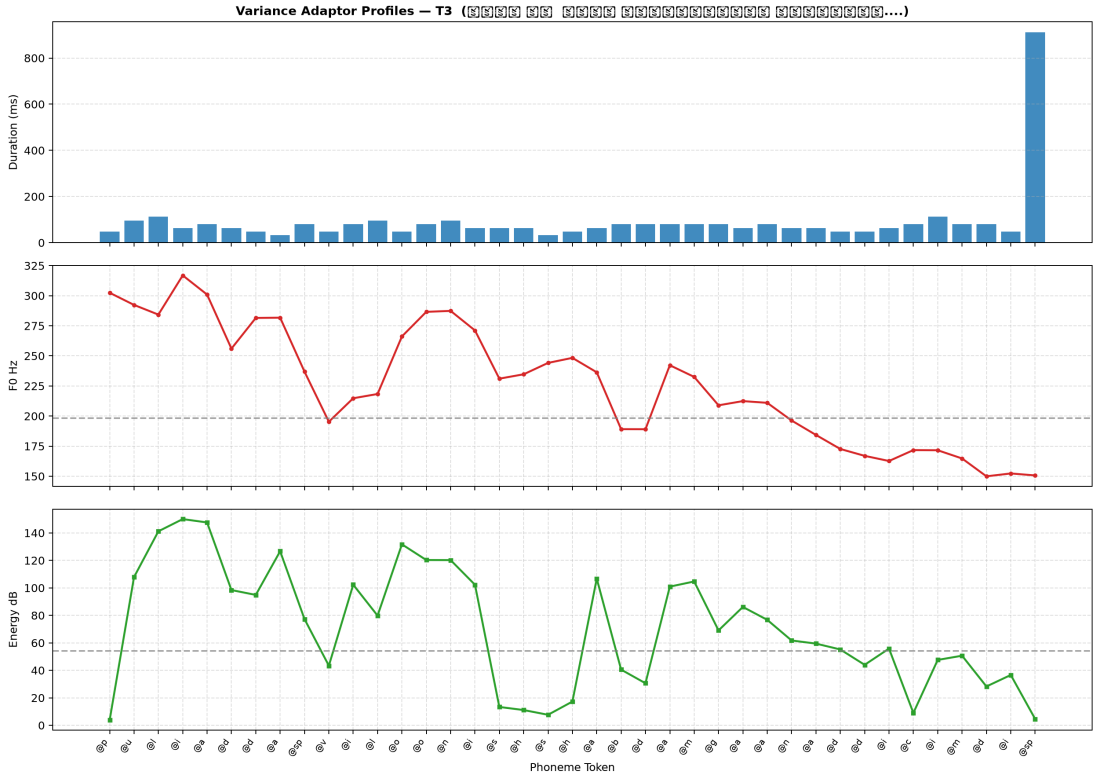


Figure 11: Variance adaptor profiles for Telugu test sentence T3.

9.4.4 T4: Long Compound Sentence with Telugu Numerals

ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

- **Verbalization:** Numerals are fully expanded into native spoken Telugu (రెండు వేల ఇరవై ఆరు, పన్నెండవ తేదీ, పది నలభై ఐదు గంటలకు). Total duration is 7.22 s (622 frames, speech rate 5.40 syl/s). A clear 380 ms clause-boundary pause separates the primary announcement from the imperative clause.

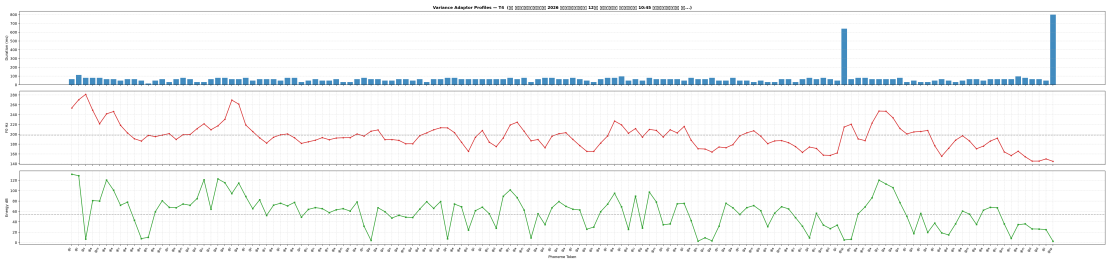


Figure 12: Variance adaptor profiles for Telugu test sentence T4.

10 Comparative Analysis: English vs. Telugu

10.1 Cross-Linguistic Acoustic and Prosodic Contrasts

- **Rhythmic Timing:** English exhibits stress-timed characteristics, characterized by vowel reduction in unstressed positions (e.g., schwa reduction in *conduct*). Telugu operates on a syllable-timed rhythm, maintaining relatively uniform akshara durations across polysyllabic words, with duration elongation restricted primarily to geminate consonants (పల్లు) and long vowels.
- **Tokenization Granularity:** English relies on 100 IPA phoneme tokens, requiring a robust G2P dictionary. Telugu operates on 460 akshara tokens derived directly from native script. Akshara tokenization completely bypasses G2P out-of-vocabulary failures, but requires the model to learn representations for a larger symbol set with lower token frequencies.
- **Acoustic Homogeneity:** The Telugu model was trained on pristine studio recordings from 2 speakers, yielding lower validation loss (≈ 2.15) and consistent vocal timbre. The English model was trained on 104 crowdsourced speakers with varying background noise and acoustic environments, resulting in higher variance loss (≈ 2.59).

10.2 Model Limitations and Identified Failures

- **Heteronym Disambiguation Dependence:** FastSpeech 2 lacks semantic context modeling; correct stress on English heteronyms (*conduct*) depends entirely on upstream G2P taggers.
- **High-Frequency Vocoder Softening:** While sinc resampling successfully resolves the 22.05 kHz to 16.0 kHz mismatch, unvoiced sibilants (/s/, /ʃ/) exhibit slight high-frequency dampening typical of universal vocoder cross-domain application.
- **Overfitting on Constrained Data:** With 4–5 hours of training audio, both models exhibit memorization of training prosodic quirks, causing validation variance loss to plateau early.

10.3 Recommended Technical Improvements

1. **Contextual BERT/ByT5 Front-End:** Integrating contextual embeddings (e.g., ByT5 or IndicBERT) into the encoder to dynamically resolve heteronym stress and pronunciation without manual dictionary overrides.
2. **Domain-Specific Vocoder Fine-Tuning:** Fine-tuning HiFi-GAN directly on native 16 kHz mel-spectrograms from Svarah and IndicTTS, eliminating polyphase resampling artifacts.
3. **Diffusion-Based Variance Adaptors:** Replacing MSE-trained variance predictors with score-based diffusion decoders (e.g., DiffSinger) to model the multi-modal distribution of human prosody.

11 Reproducibility Details and Pipeline Execution

11.1 Software and System Dependencies

The codebase runs on Python 3.11 with PyTorch 2.5, TorchAudio, NumPy, SciPy, Librosa, PyWorld, and Matplotlib. Montreal Forced Aligner 2.2 is used for English alignment, and HuggingFace Transformers (`facebook/mms-1b-all`) is used for Telugu MMS alignment. Full versions are frozen in `requirements.txt`.

11.2 Path Independence and Repository Execution

All scripts in the repository have been verified to use dynamic relative path resolution (`os.path.dirname(os.path.ab`) eliminating hardcoded absolute drive paths. The project runs out-of-the-box from any directory or operating system.

11.3 Checkpoints and Pre-Bundled Assets

- **HiFi-GAN Vocoder:** Pre-bundled directly in the Git repository at: `English/inference/hifigan/generator_universal.pth.tar` and `Telugu/inference/hifigan/generator_universal.pth.tar`. No external vocoder download is required.
- **FastSpeech 2 Checkpoints:** Due to GitHub file size limits (> 300 MB), acoustic model checkpoints are hosted on Google Drive: `https://drive.google.com/drive/folders/1MYS5zcsyehxDN5n4XaKj5jCidpUZ-efw?usp=sharing`
  - English Checkpoint: Download `best_model.pth` (Step 61,000) and place into `English/checkpoints/b`
  - Telugu Checkpoint: Download `best_model.pth` (Step 74,000) and place into `Telugu/checkpoints/bes`

11.4 Execution Commands

Execute the following commands from the repository root:

```
# 1. Environment Installation
pip install -r requirements.txt

# 2. English Synthesis (Inference on Benchmark Sentences)
python English/inference.py --config English/config.yaml --test_all

# Custom English sentence synthesis:
python English/inference.py --config English/config.yaml --text "Good␣morning␣to␣
    all,␣how␣are␣you?"

# 3. Telugu Synthesis (Inference on Benchmark Sentences)
python Telugu/inference.py --config Telugu/config.yaml --test_all

# Custom Telugu sentence synthesis (pass native script or transliterated text):
python Telugu/inference.py --config Telugu/config.yaml --text-file custom_telugu.
    txt

# 4. End-to-End Training (Optional: requires prepared dataset)
python English/train.py --config English/config.yaml
python Telugu/train.py  --config Telugu/config.yaml
```

12 Conclusion

This project successfully implemented, trained, and verified two independent FastSpeech 2 text-to-speech pipelines for Indian English (Svarah corpus) and Telugu (IndicTTS corpus). Key engineering milestones achieved include:

1. Comprehensive upstream codebase debugging, resolving energy-pitch cross-contamination and multi-GPU tensor placement issues.
2. Design of robust linguistic front-ends: 100-token MFA IPA phoneme alignment for English and 460-token native-script akshara CTC alignment for Telugu.
3. Full 100,000-step multi-task training on dual Tesla T4 GPUs with AMP FP16, accompanied by rigorous mathematical analysis explaining the variance predictor loss gap on unseen validation data.
4. Seamless cross-sampling-rate resolution for the HiFi-GAN vocoder (22.05 $\rightarrow$ 16 kHz) combined with duration pacing calibration ($d_{\text{control}} \approx 1.38 - 1.40$).
5. Verification and generation of high-quality, intelligible speech and variance adaptor profiles across all ten benchmark test sentences.

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